WORKED SOLUTION

STEP II 1990, Question 11

Mechanics • Conservation of Angular Momentum

STEP II 1990, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q11

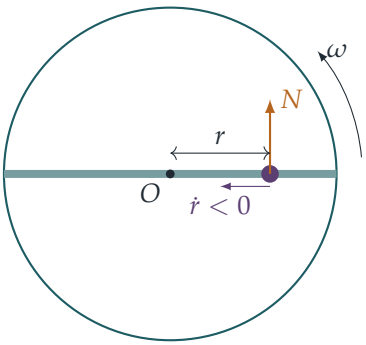
The key idea

Disc and particle together form a system with no external torque about the vertical axis and no external force doing work, so two quantities are conserved: angular momentum about the axis and kinetic energy. Those two equations give ω and $\mathrm{d}r/\mathrm{d}t$ directly. Dividing one rate by the other removes time and leaves a separable equation for the path, r as a function of θ.

Solution

The model

Treat the particle as a point mass in a smooth groove, and assume the axis is smooth (frictionless bearing). Seen from above, the particle at distance r along the groove has velocity $\dot{r}$ along the groove and $r\omega$ perpendicular to it, where $\omega = \dot{\theta}$.



The only horizontal force on the particle is the groove's reaction N, perpendicular to the groove (the groove is smooth, so there is no component along it). The disc feels $-N$.

Angular momentum

Gravity and the vertical reactions are parallel to the axis, and the smooth axis exerts no torque, so the total angular momentum about the axis is constant. The disc contributes $mk^2\omega$ and the particle $m\,r\cdot r\omega = mr^2\omega$. Comparing with $t=0$:

$$mk^2\omega + mr^2\omega = mk^2\Omega + ma^2\Omega \quad\Longrightarrow\quad \omega = \frac{\Omega(k^2+a^2)}{k^2+r^2}.$$

Energy

> **Ask yourself** The reaction N pushes the particle sideways while it moves sideways. Doesn't that change the kinetic energy?

It changes the particle's kinetic energy, but not the total. N does work on the particle at rate $N\cdot r\omega$, while the equal and opposite force on the disc, acting at the same distance r, exerts a torque $-Nr$ and so does work at rate $-Nr\omega$. These cancel, so the total kinetic energy is conserved:

$$\tfrac{1}{2}mk^2\omega^2 + \tfrac{1}{2}m(\dot r^2 + r^2\omega^2) = \tfrac{1}{2}mk^2\Omega^2 + \tfrac{1}{2}m(V^2 + a^2\Omega^2).$$

Hence

$$\dot r^2 = V^2 + (k^2+a^2)\Omega^2 - (k^2+r^2)\omega^2.$$

Using the given condition $V^2 = \Omega^2 a^2(k^2+a^2)/k^2$,

$$V^2 + (k^2+a^2)\Omega^2 = \Omega^2(k^2+a^2)\,\frac{a^2+k^2}{k^2} = \frac{\Omega^2(k^2+a^2)^2}{k^2},$$

and from the angular momentum result $(k^2+r^2)\omega^2 = \Omega^2(k^2+a^2)^2/(k^2+r^2)$. Therefore

$$\dot r^2 = \Omega^2(k^2+a^2)^2\left(\frac{1}{k^2} - \frac{1}{k^2+r^2}\right) = \frac{\Omega^2(k^2+a^2)^2\,r^2}{k^2(k^2+r^2)}.$$

While the particle moves towards the axis, $\dot r < 0$, so we take the negative square root (with $r>0$):

$$\frac{\mathrm{d}r}{\mathrm{d}t} = -\frac{\Omega r(k^2+a^2)}{k(k^2+r^2)^{1/2}},$$

as required.

The path

Since $\omega = \mathrm{d}\theta/\mathrm{d}t > 0$, the chain rule gives

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = \frac{\mathrm{d}r/\mathrm{d}t}{\mathrm{d}\theta/\mathrm{d}t} = -\frac{\Omega r(k^2+a^2)}{k(k^2+r^2)^{1/2}}\cdot\frac{k^2+r^2}{\Omega(k^2+a^2)} = -\frac{r(k^2+r^2)^{1/2}}{k},$$

that is, $k\,\mathrm{d}r/\mathrm{d}\theta = -r(k^2+r^2)^{1/2}$. Note that Ω and a have dropped out.

Now put $u = 1/r$, so $\mathrm{d}r/\mathrm{d}\theta = -u^{-2}\,\mathrm{d}u/\mathrm{d}\theta$ and $r(k^2+r^2)^{1/2} = u^{-2}(1+k^2u^2)^{1/2}$ (as $u > 0$). The equation becomes

$$k\frac{\mathrm{d}u}{\mathrm{d}\theta} = \sqrt{1+k^2u^2}.$$

The square root suggests a hyperbolic substitution: let $ku = \sinh\phi$. Then $k\,\mathrm{d}u/\mathrm{d}\theta = \cosh\phi\,\mathrm{d}\phi/\mathrm{d}\theta$ and $\sqrt{1+\sinh^2\phi} = \cosh\phi$, so

$$\frac{\mathrm{d}\phi}{\mathrm{d}\theta} = 1 \quad\Longrightarrow\quad \phi = \theta + \alpha.$$

At $\theta = 0$ we have $r = a$, so $\sinh\alpha = ku = k/a$. Hence $k/r = \sinh(\theta+\alpha)$, i.e.

$$r\sinh(\theta+\alpha) = k, \qquad \sinh\alpha = \frac{k}{a}.$$

The particle never reaches the axis

> **Ask yourself** $r = k/\sinh(\theta+\alpha)$ tends to 0 as $\theta \to \infty$. Could the disc turn through an infinite angle in a finite time?

No. Since $r^2 \geq 0$, $\omega = \Omega(k^2+a^2)/(k^2+r^2) \leq \Omega(k^2+a^2)/k^2$, so by time t the disc has turned through at most $\theta \leq \Omega(k^2+a^2)t/k^2$, which is finite. For any finite θ, $\sinh(\theta+\alpha)$ is finite, so $r = k/\sinh(\theta+\alpha) > 0$. Also $\dot{r}$ is strictly negative for every $r > 0$, so the particle never stops and turns back: the formulae stay valid for all t, and r decreases towards 0 without ever reaching it.

(Alternatively, straight from the $\mathrm{d}r/\mathrm{d}t$ equation: the time to go from a to r is $\frac{k}{\Omega(k^2+a^2)}\int_r^a \frac{\sqrt{k^2+x^2}}{x}\,\mathrm{d}x$, and near $x = 0$ the integrand behaves like k/x, whose integral diverges logarithmically as $r \to 0$.)

Checks

- At $t = 0$, $r = a$ gives $\omega = \Omega$ and $\dot{r} = -\Omega a\sqrt{k^2+a^2}/k = -V$, matching the initial data; and $\theta = 0$ gives $r = k/\sinh\alpha = a$.
- For large θ, $r \approx 2k\mathrm{e}^{-(\theta+\alpha)}$: the particle spirals in exponentially, consistent with never arriving.
- Numerically, stepping the equations of motion ($\ddot{r} = r\omega^2$ with angular momentum fixed) for random k, a, Ω reproduced $\dot{r}$, the energy and $r\sinh(\theta+\alpha) = k$ to about 10^{-12}. With $k = a = \Omega = 1$, the time to reach $r = 10^{-8}$ is about 9.32, growing like $\frac{1}{2}\ln(1/r)$.

Common slips

- Assuming the particle's energy alone is conserved: the groove's reaction does work on it, and only the total is conserved.
- Forgetting the transverse velocity $r\omega$ in the particle's kinetic energy or angular momentum.
- Concluding the particle never reaches the axis from $r > 0$ for finite θ, without checking that θ stays finite in finite time.

Beyond STEP

The special condition on V puts the particle exactly on the trajectory that creeps towards $r = 0$ for ever: in the rotating frame $r = 0$ is an unstable equilibrium, and this is its *stable manifold*. Any slightly larger V sends the particle through the centre, any smaller one makes it turn back. Ideas like this, along with conservation laws for coupled bodies, appear in Kleppner and Kolenkow, *An Introduction to Mechanics*.

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